Graphical Abstract

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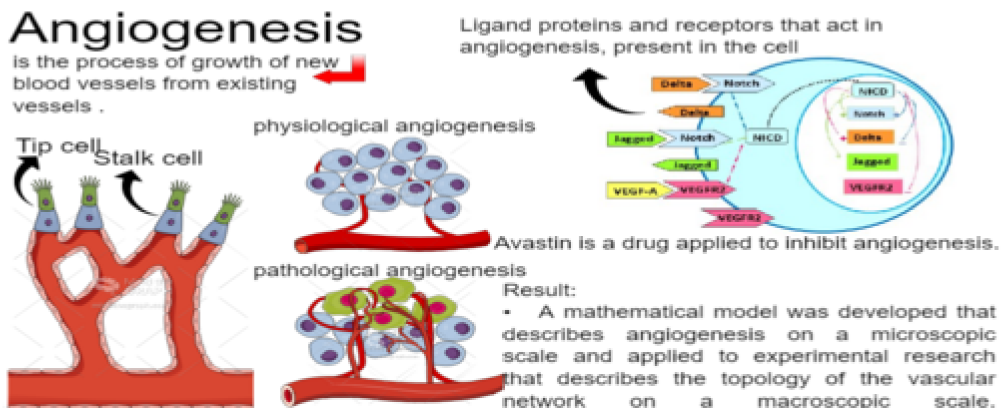

Highlights

A mathematical model describing the tip-stalk regulation in angiogenesis

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- The mathematical model satisfactorily represents the concentrations of receptor proteins and ligands involved in angiogenesis
- The model describes angiogenesis on a microscopic scale and has been applied to experimental research describing vascular network topology on a macroscopic scale
- The exclusion of *fringe* does not promote changes in the characterization of the equilibrium points of the dynamic system
- The balance of the system is disturbed when the concentration of VEGF is changed
- The decrease in VEGF-A induces the unstable system to move towards stability

A mathematical model describing the tip-stalk regulation in angiogenesis

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Abstract

Angiogenesis is the process of growth of new blood vessels from existing vessels that involves a large number of cell signalization. Under normal conditions, the new vessels are robust and organized; there is a balance among the angiogenesis factors. In abnormal conditions, as occurs in the emergence of a tumor, the vessels formed are stunted and tangled; and in this condition, there is no balance among the angiogenesis factors. In pathological angiogenesis, blood vessels are stimulated to grow rapidly to feed the tumor that lacks oxygen and nutrients. The attempt to inhibit angiogenesis can cause several side effects such as hypertension, thrombosis, fatigue, among others. To better understand the mechanism that drives angiogenesis a large effort has been done to characterize the signaling pathways that activate this process, the results from this research contribute to works aimed at the use of new drugs for various diseases, including cancer. The main aim of this study is to present a mathematical model that describes angiogenesis on a micro-

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scopic scale and apply it to experimental research that describes the topology of the vascular network on a macroscopic scale. The adapted mathematical model is described by ordinary differential equations in MatLab® software which describe the behavior of a single cell. By applying the model to experimental research, the results point out that the balance of the system is disturbed when the concentration VEGF (Vascular Endothelial Growth Factor) is altered, promoting changes in the angiogenesis process.

Keywords:
Angiogenesis, Physiological Angiogenesis, Pathological Angiogenesis, Protein Dynamics, Mathematical Modeling

1. Introduction

Angiogenesis is the process that involves the growth, migration, and differentiation of endothelial cells (ECs) [1, 2] . This mechanism is important for wound healing, embryo formation, to direct oxygen and nutrients to cells, among other processes, and is associated with several growth factors [3, 4].

During angiogenesis, endothelial cells, which are the cells that cover the inner part of blood vessels, lead the growth of new branches [5]. These cells are guided through the extracellular matrix (substance between cells) by chemicals that are released in response to nutrient deficiencies in tissues [5, 6].

The vascular network formation involves the interaction of cell-cell and cell-matrix extracellular, understanding the chemical mechanism that drives the process and displays different topologies in the vascular network plays an important role to investigate the angiogenesis process.

However, angiogenesis is also related to the emergence of several diseases, as well as the emergence of tumors [7]. Thus, tumor angiogenesis is essential for the development of cancer, acting during growth, progression and metastasis [8, 7].

Studies on the action of angiogenesis in the development of cancer began in the year 1800. On this occasion, German researchers observed that some tumors were richly vascularized, suggesting that the formation of new blood vessels occurred in some cancers [9].

In this sense, it was discovered that tumor cells also promote angiogenesis by emitting biological signals that indicate hypoxia [10]. Therefore, research that seeks to identify pro-angiogenic factors and methods of restricting signaling is of great relevance.

Under normal conditions, new blood vessels are formed in a robust and organized way. And in abnormal conditions, such as when a tumor appears, the vessels formed are excessive, rickety, and tangled. These abnormal conditions lead to the creation of a network of vessels with tight and compromised junctions, as they are stimulated to grow rapidly to feed the tumor that lacks oxygen and nutrients [11, 12]. To understand this change, it is necessary to carefully analyze the conditions that contribute to the accelerated growth of these vessels, in this sense, it is important to study the stimulating factors of angiogenesis.

To investigate the dynamics that drive the angiogenesis process on a microscopic scale, a mathematical model of angiogenesis regulation will be presented, based on the models of Boareto and collaborators [5, 13]. This model will be applied to the experimental study carried out by Alves [14], which proposes to understand the changes promoted in the topology of the vascular network through the restriction of VEGF during the process of vasculogenesis. Therefore, the concentrations of proteins involved in this process will be analyzed to evaluate the possible influence of angiogenic growth factors.

The two signaling pathways acting in angiogenesis will be addressed, which are the cellular and extracellular communication processes: the *Notch* signaling pathway and the VEGF (*Vascular Endothelial Growth Factor*) signaling pathway. The *Notch* signaling pathway, named after the *Notch* receptor protein, is involved in choosing which cells will lead the new branch of the blood vessel and which cells will form the walls of the new vessel. The VEGF signaling pathway, which takes its name from the signaling factor released by cells that lack oxygen and nutrients, emits pro-angiogenic factors for the growth of these new blood vessels.

The restriction of pro-angiogenic factors has been the subject of studies over the years [15, 14]. However, studies show that attempts to inhibit angiogenesis can cause several side effects in patients, such as high blood pressure, heart failure, arterial thrombotic events, nausea, anxiety, intense pain and sleep disorders [16, 17].

Therefore, research that focuses on describing the angiogenesis process helps investigations aimed at the use of new medicines for various diseases, including cancer. Therefore, this study is necessary, as it will provide quantitative and qualitative data to describe the cellular dynamics involved in angiogenesis. Furthermore, these data may contribute in the future to research that investigates new therapies that minimize the risks and side effects in the treatment of cancer and other diseases.

Several studies, such as those by Leite [18], Domingues [19], Silva [20], Boareto and others [5, 13], describe the action of tumor angiogenesis through mathematical and computational modeling. This approach makes it possible to effectively understand and characterize biological systems.

According to Freire [21], computational mathematical models that consider the specific biological variables of the system adequately simulate biological processes faithfully.

Thus, mathematical and computational modeling proves to be effective for investigating the equilibrium state of a dynamic system that describes angiogenesis. Furthermore, the dynamics of the action of the system's proteins can be assessed through modeling. This dynamics, described through first-order differential equations, enables understanding when manipulating protein concentrations in the simulation of virtual cells and tissues. This process provides analyses, conjectures, inferences and conclusions about biological functioning.

Therefore, a mathematical model for angiogenesis will be presented, based on first-order differential equations, which describes the dynamics of proteins considering a single cell in the middle of an external environment in which the different concentrations present are simulated.

2. *Notch* and VEGF signaling pathways

One of the receptor proteins that act in the angiogenesis process is *Notch*. *Notch* is a transmembrane protein (which covers the cell membrane from one side to the other) that plays an important role in [22] cell fate. Its signaling pathway is one of the most investigated due to its versatility in the cellular communication process [23, 24, 25].

The *Notch* signaling pathway has a direct route between the membrane and the cell nucleus and presents cell-cell communication. In this process, transmembrane binding proteins from an external environment activate the cell's receptor proteins [26].

Around 110 years ago, studies on *Notch* began and today it is known that *Notch* signaling participates in several biological processes between species, such as organ formation and tissue function. Therefore, the malfunction of this signaling may indicate a pathological environment [27].

Some studies show that the *Notch* signaling pathway is much more extensive and complex than previously believed [28, 29]. As immunotherapy has revolutionized cancer treatment, the *Notch* signaling pathway and its relationship with angiogenesis has attracted the attention of scientists [27].

In this sense, there is also the binding protein *Delta* that acts in the angiogenesis process. *Delta* has been identified as a potentially important target in tumor angiogenesis [30]. Studies on tumors in mice and humans have shown that *Delta* is strongly expressed in tumor angiogenesis compared to physiological angiogenesis [31].

Thus, the binding between *Notch* and *Delta* allows the NICD to migrate to the nucleus of the cell containing *Notch* of the binding and restrict the rate of production of new *Delta* proteins, but increase the rate of protein production *Notch* [31].

Notch-Delta signaling causes neighboring cells to acquire opposite patterns. That is, a cell with high *Delta* and low *Notch* promotes that its neighboring cells have high *Notch* and low *Delta* on its surface. This pattern is called lateral inhibition and ensures that cells differentiate into distinct fates from an initially homogeneous cell population [32, 33]. By definition, cells with high *Delta* are called emitters and cells with high *Notch* are called receivers [13].

In addition to the *Delta* ligand, there is the *Jagged* ligand protein that also acts in the angiogenesis process. Research reveals that the excessive increase in *Jagged* is related to tumor angiogenesis, as well as some types of cancer such as breast and ovarian [34, 27]. The study carried out in 1997 and presented in [35] shows that the *Jagged* mutation is also related to the development of Alagille Syndrome, which is an autosomal dominant disease characterized by abnormal development of organs.

In this sense, the connection between *Notch* and *Jagged* allows the NICD to migrate to the nucleus of the cell that contains *Notch* of the connection and increase the production rates of *Notch* and *Jagged* simultaneously [5, 36].

Notch-Jagged signaling promotes neighboring cells to acquire similar patterns. That is, cells with high *Jagged* and high *Notch*, being called Sender/ Receiver hybrid phenotypes, to highlight that both cells send and receive signals [13]. This pattern is called lateral induction and guarantees similar cell fates [36].

If the connection between *Notch* and *Delta*, or between *Notch* and *Jagged*, with between different cells, there is a *trans* activation. However, if it occurs in the same cell, there is *cis* inhibition and, in this case, the NICD is not activated and both proteins are degraded in the process [5, 13].

In addition to the characteristics presented here, [5] consider another factor called the *fringe* effect, which divides the *Notch* population into two parts, making one of them more susceptible to receiving *Delta* instead of *Jagged*.

In the *Notch* signaling pathway, *fringe* acts to coordinate binary cell fate decisions during angiogenesis [37]. Furthermore, growing evidence has highlighted *fringe*'s role in cancer, suggesting that its modulation may reduce cancer cell proliferation [38].

In this sense, cells with a high concentration of *Jagged* emit signals that suppress the activation of *Notch* modified by *fringe*, so that the expression *Notch-Delta* is reinforced [37]. In this scenario, loss of regulation of the pathway by *Notch-Jagged* signal suppressed by *fringe* causes destabilization of cell fate *tip* [37].

Another very important signaling pathway in the angiogenesis process is VEGF (*Vascular Endothelial Growth Factor*). Tumor angiogenesis has become important for researchers aiming to develop antitumor therapies, with most therapies aimed at blocking the VEGF signaling pathway [31].

VEGF family proteins are regulators of angiogenesis, both in physiological circumstances and in a pathological condition [31]. Thus, there is the VEGF-A binding protein, also named simply as VEGF, and the VEGFR-2 receptor protein, also named as VEGFR2 or VEGF-2 [39].

In VEGF signaling, the VEGF-A ligand, found on the outside of the cell, activates the VEGFR2 receptor, stimulating angiogenesis. Although VEGF is essential to maintain the relative condition of stability in cells and tissues, the importance of this factor for tumor growth and metastasis dissemination has been demonstrated [12]. Furthermore, [5, 13] showed a great influence of the *Jagged* protein in the *Notch* signaling for the appearance of tumors.

In the process of new blood vessel growth, the endothelial cell that leads the growth of the new vessel is called a *tip* cell, and new thin extensions (phyllopods) are created in this cell to follow the signalings emitted by VEGF-A. The remaining cells that follow the *tip* cell proliferate and form the stem of the new blood vessel. Those are the *stalk* cells. As has been shown the mechanical regulation of cells in angiogenesis process gives rise to the mean of chemical transduction [40].

Tissues rich in proteins *Delta* and VEGFR2 give rise to *tip* cells and, tissues rich in proteins *Jagged*, NICD and *Notch*, give rise to cells *stalk*. The vessels emerge using a well-regulated balance between the migration of *tip* cells and the proliferation of *stalk* cells [41]. As has been shown the Jagged plays an important role in tumor progression [42]. To understand the vascular network formation a large number of mathematical models have been proposed to study the cell's interactions [43]. On the other hand, there are many factors involved in the microscopic scale that drives the behavior in the macroscopic scale that still not understand. So, this work proposes a model that describes how the possible single-cell interactions give rise to the macroscopic behavior in the angiogenesis process.

The high concentration of *Delta* and VEGFR2 in the *tip* cell causes its neighbors to acquire an opposite pattern (with low *Delta* and VEGFR2) called lateral inhibition. On the other hand, the high concentration of *Jagged*, NICD, and *Notch* in the *stalk* cell causes its neighbors to acquire a similar pattern, called lateral induction, which will promote vessel elongation. The interaction between the *VEGF* and *Notch* signaling pathways is important to define the fate of *tip* and *stalk* cells.

Studies show that composite cells with *tip/stalk* characteristics promote the emergence of tumor angiogenesis, as there is competition between the position of guiding new vessels towards VEGF-A [5, 44]. This causes blood vessels to grow faster, poorly perfused and disorganized.

3. Vascular network topology

The study presented by Alves [14] analyzes the formation of the vascular network in chicken embryos of the species gallus gallus domesticus. The embryos were divided equally into two distinct groups: the control group (used to extract quantitative parameters) and the group treated with $30\mu L$ of Avastin® (*Bevacizumab* or *Bevacizumabe*) at a concentration of $6\mu g/\mu L$ (used to verify the influence of VEGF on vascular network development).

Avastin® is an antibody used in chemotherapy treatments and inhibits the action of VEGF. According to Susil Kumar Mukerjee [45], Avastin® is applied to inhibit angiogenesis, and its use in early-stage tumors has been studied over the years. Before that, Avastin® was widely used to treat metastases.

When Avastin® is added to the system, it receives the VEGF-A ligand contributing so that the signaling between VEGFR2 and VEGF-A is not activated and, consequently, angiogenesis is slowed down. When there is no influence of the drug on the system, the link between VEGFR2 and VEGF-A forms the VEGFR + VEGF-A set, which enables the activation of angiogenesis [14].

The main result presented by Ana Alves [14] is that the network topologies obtained for the two groups of embryos showed differences. In the control group, VEGF binds to molecules presents in the extracellular matrix and enables the formation of pathways that stimulate the migration of endothelial cells, forming a tree-like vascularization. In samples treated with Avastin®, it is observed that a symmetry break of the bonds occurs [14]. This is an indication that VEGF-A, present in the extracellular matrix, binds to Avastin® and the bindings in this type of sample do not follow the same paths as in the control group. This statement corroborates what was seen in the descriptions presented in [5], as VEGF induces the formation of new shoots and the cell *tip* migrates towards the signal emitted by VEGF leading to a new angiogenic branch. Furthermore, studies suggest that tumors treated with antiangiogenic drugs have their vascular network reorganized to improve oxygenation and drug penetration into the system [46].

Through a complete mechanism, several factors influence the promotion or inhibition of angiogenesis [47]. Therefore, theoretical work seeks to understand how the dynamics of blood vessel formation occur. Theoretical studies such as those presented in [5, 13] investigate the microscopic mechanisms of cellular chemistry involved in the growth of the vessels that feed a tumor. These researches can contribute to the search for less invasive drugs that focus on restricting the processes that, in tumors, lead to the unrestrained proliferation of blood vessels [6].

4. Material and Methods

Initially, a detailed study was carried out on the mathematical models, which correspond to angiogenesis,developed by Boareto and collaborators [5, 13]. Contrary to what was done by these authors, in the mathematical model developed in this research, the *fringe* effect was not considered as a way of verifying whether it influences the results obtained, which will be compared with those presented in [5] and [13].

The mathematical model obtained was applied to the experimental study of Alves [14]. From the action performed by VEGF-A in angiogenesis, these results were associated with the results regarding the formation of the topology of the vascular network presented in [14].

All mathematical models were built using first-order differential equations when considering the dynamics of the *Notch* and VEGF signaling pathways.

To perform the computational analyses, the data processing was done using MatLab® software.

To validate the model, the analyzes performed were compared with the investigations carried out in [5] and [13] and reproduced the same qualitative behavior, even not considering the *fringe* effect in the model as proposed here.

5. Results and Discussions

The signaling pathway *Notch* is activated when the receiver *Notch* receives the binders *Delta* or *Jagged*. When *Notch* receives *Delta*, the NICD is activated and, in the nucleus of the cell containing *Notch* the production rate of *Notch* is increased, but the production rate of *Delta* is restricted. When *Notch* receives *Jagged* the NICD is also activated and in the cell nucleus causes both *Notch* and *Jagged* to have their production rates increased. When the receptor and ligands interact between different media, or different cells, the NICD is activated and this dynamic is called trans activation. However, when the receptor and ligands interact in the same cell, the NICD is not activated, the proteins involved in the process are degraded and this dynamic is called cis inhibition [5, 13] (Figure 1).

Moreover, there is also the VEGFR2 receptor and the VEGF-A ligand. When VEGFR2 receives VEGF-A, the NICD is activated and in the cell nucleus causes the production rate of *Delta* to be increased and the production rate of VEGFR2 to be restricted [5, 13] (Figure 1) .

Considering this dynamic, it is noteworthy that cells with high concentrations of *Delta* and VEGFR2 and with low concentrations of NICD and *Jagged* are called *tip* cells, which are cells that migrate towards VEGF-A signaling, thus guiding the formation of new blood vessels. However, cells with high concentrations of NICD and *Jagged* and with low concentrations of *Delta* and VEGFR2 are called stem cells, which form the wall of new blood vessels [5].

The mathematical model that will be presented is adapted from [5]. In the model presented in [5] a phenomenon called the *fringe* effect is considered. It is an effect that modifies part of the *Notch* population to make it more capable of receiving *Delta* instead of *Jagged*. Contrary to what was done in [5], the The *fringe* effect will not be considered in the mathematical

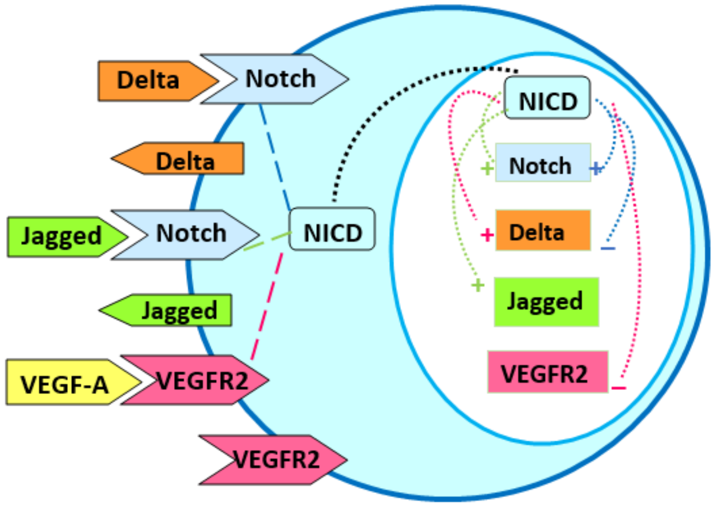

Figure 1: Representation of a cell interacting with the external environment. There are ligands and receptor proteins on the cell membrane and in its nucleus. *Notch* receives *Delta* and *Jagged* from an external environment. This binding activates the NICD, which migrates to the cell nucleus and increases the production of new *Notch* and *Jagged* proteins, but restricts the production of *Delta*. VEGFR2 receives VEGF-A. Activated NICD migrates to the nucleus and increases the production of *Delta*, but restricts the production of VEGFR2.

model developed in this research, in order to evaluate its possible qualitative influence on the system, and, for this, the results of the system dynamics will be compared with the results presented in [5].

Therefore, when considering the dynamics described and presented in Figure 1, the mathematical model adapted from Boareto and collaborators [5] is represented by the following equations

$$\frac{dN}{dt} = N_0 H^{S+}(I, \lambda_{I,N}) - k_t N(D_{ext} + J_{ext}) - k_c N(D + J) - N \tag{1}$$

$$\frac{dD}{dt} = D_0 H^{S-}(I, \lambda_{I,D}) H^{S+}(V, \lambda_{V,D}) - k_t N_{ext} D - k_c N D - \gamma D \tag{2}$$

$$\frac{dI}{dt} = k_t N(D_{ext} + J_{ext}) - \gamma_I I \tag{3}$$

$$\frac{dJ}{dt} = J_0 H^{S+}(I, \lambda_{I,J}) - k_t N_{ext} J - k_c N J - \gamma J \tag{4}$$

$$\frac{dV}{dt} = k_t V_{ext} VR - \gamma_I V \tag{5}$$

$$\frac{dVR}{dt} = VR_0 H^{S-}(I, \lambda_{I,VR}) - k_t V_{ext} VR - \gamma VR \tag{6}$$

where N, D, I, J, V, and VR represent, respectively, the concentrations of the proteins *Notch*, *Delta*, NICD, *Jagged*, VEGF and VEGFR2. N_0, D_0, I_0, J_0 and VR_0 represent, respectively, the production rates of *Notch*, *Delta*, NICD, *Jagged* and VEGFR2 which are enlarged or restricted in the cell nucleus. The term k_t represents the trans interaction and the term k_c represents the cis inhibition. N_{ext}, D_{ext}, J_{ext} and V_{ext} represent, respectively, the external cell concentrations of the proteins *Notch*, *Delta*, *Jagged* and VEGF (which is VEGF-A). Also, γ and γ_I are the natural degradation rates of proteins. The term $H^{S+}(I, \lambda_{I,N})$ represents the shifted Hill equation, which causes the production rate of *Notch* to be increased in the kernel, due to the action of the NICD. In a similar way, $H^{S-}(I, \lambda_{I,D})$ represents the shifted Hill equation that constrains the production of *Delta* in the kernel under the action of the NICD and $H^{S+}(V, \lambda_{V,D})$ represents the increase of *Delta* in the cell nucleus via the VEGF signaling pathway. The term $H^{S+}(I, \lambda_{I,J})$ represents the shifted Hill equation that regulates the increase of *Jagged* in the kernel through the action of the NICD. The shifted Hill equation that constrains the production rate of VEGFR2 is given by $H^{S-}(I, \lambda_{I,VR})$. For details on constructing the model equations, see Appendix A.

Therefore, the model is composed of six equations, each of which represents the dynamics of a protein present in the angiogenesis process.

It is noteworthy that the model equations describe the concentrations present in a single cell, which is immersed in an external environment where the other external concentrations are simulated.

5.1. Model Equilibrium Analysis

Using the model equations and the parameters presented in Appendix A.2, Figure 2 presents the dynamics of the system's proteins and the nullclines without the influence of VEGF.

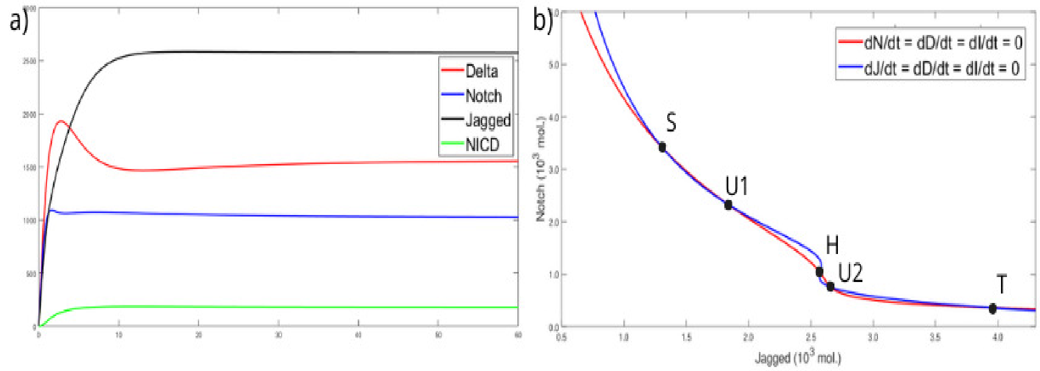

Figure 2: Dynamics and equilibrium points of the model in the absence of VEGF. a) Signaling growth behavior in the absence of VEGF. It is observed that the system has a greater quantity of ligands proteins (*Jagged* and *Delta*) and, consequently, a smaller quantity of receptor proteins (*Notch* and NICD). b) Nullclines. The meeting points between the curves show the existence of five equilibrium points, called S, U1, H, U2 and T.

Figure 2a) presents the concentrations of values reached by proteins of the Notch signaling pathway, without the influence of VEGF. It is observed that the production of the binding protein *Jagged* is the highest in the system. Furthermore, the concentration of *Delta* binding proteins is greater than that of the *Notch* receptor and NICD. This reflects a state of high concentration of binding proteins. It should be noted that this dynamics is obtained for a particular set of conditions for N_0, D_0, J_0 and I_0.

Figure 2b) shows Notch versus Jagged and the existence of five equilibrium points is observed, which are the points where the two curves intersect. These equilibrium points will be called $S, U1, H, U2$ and T. Below, $\vec{x_1}, \vec{x_2}, \vec{x_3}, \vec{x_4}, \vec{x_5}$ present the concentrations of each protein in $S, U1, H, U2$ and T, respectively.

$$\vec{x_1} = \begin{bmatrix} N \\ D \\ J \\ I \end{bmatrix} = \begin{bmatrix} 3385,2 \\ 101,3 \\ 1317,5 \\ 592,4 \end{bmatrix}, \vec{x_2} = \begin{bmatrix} N \\ D \\ J \\ I \end{bmatrix} = \begin{bmatrix} 2329,2 \\ 270,8 \\ 1835,3 \\ 407,6 \end{bmatrix}, \vec{x_3} = \begin{bmatrix} N \\ D \\ J \\ I \end{bmatrix} = \begin{bmatrix} 1025,7 \\ 1562,9 \\ 2573,7 \\ 179,5 \end{bmatrix}$$

$$\vec{x_4} = \begin{bmatrix} N \\ D \\ J \\ I \end{bmatrix} = \begin{bmatrix} 751,1 \\ 2511,4 \\ 2659,2 \\ 131,4 \end{bmatrix}, \vec{x_5} = \begin{bmatrix} N \\ D \\ J \\ I \end{bmatrix} = \begin{bmatrix} 360,2 \\ 5367,2 \\ 3945,6 \\ 63 \end{bmatrix}$$

Now, the stability of points $S, U1, H, U2$ and T, presented in Figure 2b) will be mathematically evaluated. According to Sayama [48], after finding the equilibrium points, the next step to analyze the linear stability of continuous-time nonlinear systems is to calculate the Jacobian matrix. When using the parameters presented in Appendix A.2 and calculating the partial derivatives to compose the Jacobian matrix J, we have

$$J = \begin{bmatrix} -\frac{\mathrm{D}}{2*10^3} - \frac{J}{2*10^3} - \frac{3}{16} & -\frac{N}{2*10^3} & -\frac{N}{2*10^3} & \frac{4\,\mathrm{I}}{25\,\sigma_2} - \frac{2\,\mathrm{I}}{25\,{\sigma_2}^2} - \frac{\mathrm{I}^3}{25*10^4\,{\sigma_2}^2} \\ -\frac{\mathrm{D}}{2*10^3} & \sigma_3 & 0 & -\frac{9\,\mathrm{I}}{10^2\,{\sigma_2}^2} \\ -\frac{J}{2*10^3} & 0 & \sigma_3 & \frac{\sigma_4}{8*10^7\,\sigma_1} - \frac{\sigma_4}{16*10^7\,{\sigma_1}^2} - \frac{3\,\mathrm{I}^9}{256*10^17\,{\sigma_1}^2} \\ \frac{7}{80} & 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

Where $\sigma_1 = \frac{\mathrm{I}^5}{32*10^{10}} + 1, \sigma_2 = \frac{\mathrm{I}^2}{4*10^4} + 1, \sigma_3 = -\frac{N}{2*10^3} - \frac{1}{8}$ and $\sigma_4 = 3\,\mathrm{I}^4$

After substituting the equilibrium values, presented by $\vec{x_1}, \vec{x_2}, \vec{x_3}, \vec{x_4}$ and $\vec{x_5}$, in the Jacobian matrix, they were found the following eigenvalues

$$\vec{x_1}(\lambda) = \begin{bmatrix} -2,5365 \\ -0,0367 \\ -0,6413 \\ -1,8176 \end{bmatrix}, \vec{x_2}(\lambda) = \begin{bmatrix} -2,3767 \\ 0,0321 \\ -0,6855 \\ -1,2896 \end{bmatrix}, \vec{x_3}(\lambda) = \begin{bmatrix} -2,9067 \\ -0,0556 \\ -0,4314 \\ -0,6379 \end{bmatrix}$$

$$\vec{x_4}(\lambda) = \begin{bmatrix} -3,7782 \\ 0,0344 \\ -0,5296 \\ -0,5005 \end{bmatrix} \vec{x_5}(\lambda) = \begin{bmatrix} -5,0952 \\ -0,0596 \\ -0,4991 \\ -0,3051 \end{bmatrix}$$

The last step refers to the analysis of stability according to the eigenvalues obtained.

In this sense, the dominant eigenvalue of $\vec{x_1}(\lambda)$ is $-0.0367 < 0$, of $\vec{x_3}(\lambda)$ is $-0.0556 < 0$ and of $\vec{x_5}(\lambda)$ is $-0.0596 < 0$. Therefore, these are stable equilibrium points. This means that even when adding a small perturbation in the vicinity of each of these equilibrium points, the protein concentrations tend to return to the equilibrium values given in $\vec{x_1}, \vec{x_3}, \vec{x_5}$.

In $\vec{x_2}(\lambda)$ the dominant eigenvalue is $0.0321 > 0$ and in $\vec{x_4}(\lambda)$ the dominant eigenvalue is $0.0344 > 0$. In this case, the points equilibrium points are unstable and, furthermore, as the other eigenvalues are less than zero, these equilibrium points are saddle points. As equilibria are unstable, this means that when adding a small perturbation in the vicinity of these points, the system tends to move infinitely away from the concentrations presented in $\vec{x_2}$ and in $\vec{x_4}$. However, an unstable equilibrium point can come with other eigenvalues that show stability, as is this case. At the saddle point, nearby trajectories are attracted to the equilibrium point in some directions, but are repelled in other directions [48].

When checking the concentrations of proteins present at the stable equilibrium point S, high concentrations of *Notch* and Jagged proteins were found. Furthermore, low concentrations of *Delta*. Therefore, point S has characteristics of *stalk* cells, which are the cells that will compose and lengthen the walls of blood vessels, corroborating the results obtained in [5].

When analyzing the values achieved for each protein at point T, it was observed that the stable equilibrium point T has a high concentration of protein *Delta* and a low concentration of *Notch*. Therefore, it can be concluded that this point characterizes *tip* cells, which are the cells that will guide the growth of new blood vessels, corroborating the results obtained in [5].

Finally, when analyzing the concentrations at point H, a new state for the system was discovered. The asymmetry of the NICD in restricting *Delta* and increasing *Jagged* production is responsible for the emergence of this hybrid stable equilibrium state, which causes cells to acquire similar fates , being receptors and ligands at the same time [13]. The analyzes showed that for this point there are intermediate concentrations of the *Notch* repeater and the *Delta* and *Jagged* ligands, as found in [13]. Therefore, this stable equilibrium point with hybrid concentrations allows the cell to emit and receive signals, having characteristics of receptor and ligand at the same time [13]. In this hybrid state, cells will have *tip* and *stalk* characteristics at the same time, which can lead to a pathological state of angiogenesis, where their states compete for the position of guiding the growth of new blood vessels to supply the body of oxygen.

Introducing the VEGF signaling pathway to the analyses, using the model equations, and the parameters presented in Appendix A.2, Figure 3 presents the dynamics and equilibrium points of the system.

Figure 3a) shows the concentrations of values reached by proteins from the Notch and VEGF signaling pathways. It is observed that the concentration of *Notch* receptor protein is predominant in the model. Furthermore, the concentrations of *Jagged* and NICD are predominant in relation to the

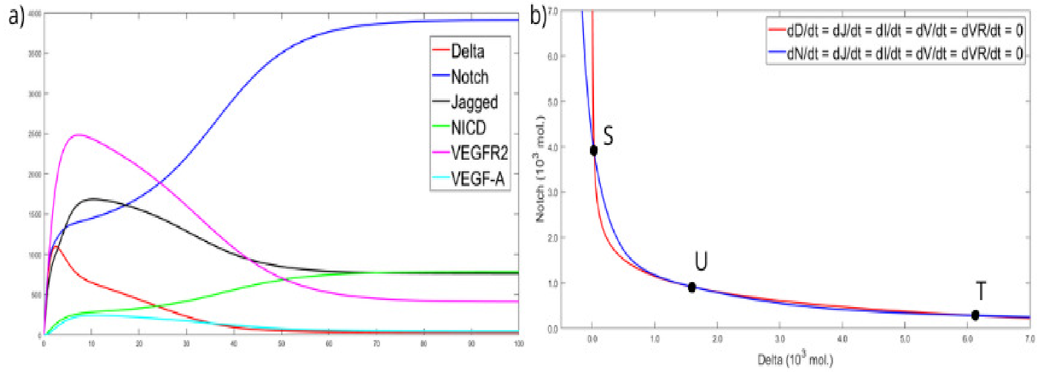

Figure 3: Dynamics and equilibrium points of the model. a) Signaling growth behavior in the presence of VEGF. The significant concentration of *Notch* in the system and the intermediate concentrations of NICD and *Jagged* are observed. Furthermore, the system presents lower concentrations of *Delta*, VEGFR2 and VEGF-A proteins. b) Nullclines. The meeting points between the curves show the existence of three equilibrium points: one unstable (U) and two stable (S and T).

concentrations of VEGFR2, *Delta* and VEGF. Given the specific parameters used, this dynamic represents a state of concentration of *stalk* cells, which are the cells that lengthen the formation of the walls of new blood vessels It can be concluded that the model reproduces the expected behavior, that is, NICD favors the production of *Notch* and *Jagged*, however *Notch* stabilizes with a higher value than *Jagged* because the production rates of these proteins are different and $N_0 > J_0$. On the other hand, NICD inhibits the production of *Delta* which could be favored if the VEGF concentration was high; however, VEGF growth is also inhibited by NICD.

It should be noted that this result reflects behavior for a particular set of conditions for N_0, D_0, J_0, I_0 and VR_0 and in it, a high concentration of *Notch* is noted, which represents a state of cells *stalk*.

Figure 3b) shows *Notch* versus *Delta* and the existence of three equilibrium points is observed, which are the points where the two curves intersect. These equilibrium points will be called S, U and T. Below, $\vec{x_1}, \vec{x_2}, \vec{x_3}$ present the concentrations of each protein in S, U and T, respectively.

$$\vec{x_1} = \begin{bmatrix} N \\ D \\ J \\ I \\ VR \\ V \end{bmatrix} = \begin{bmatrix} 274,2 \\ 6173,5 \\ 2790,7 \\ 54,8 \\ 620 \\ 6200,4 \end{bmatrix}, \vec{x_2} = \begin{bmatrix} N \\ D \\ J \\ I \\ VR \\ V \end{bmatrix} = \begin{bmatrix} 900,9 \\ 1628,8 \\ 1828,5 \\ 180,2 \\ 368 \\ 3679,8 \end{bmatrix}, \vec{x_3} = \begin{bmatrix} N \\ D \\ J \\ I \\ VR \\ V \end{bmatrix} = \begin{bmatrix} 3909 \\ 30,4 \\ 759,9 \\ 781,8 \\ 40,9 \\ 409,5 \end{bmatrix}$$

Now, the stability of points S, U and T, presented in Figure 3b) will be evaluated mathematically. When using the parameters presented in Appendix A.2 and calculating the partial derivatives to compose the Jacobian matrix J, we have

$$J = \begin{bmatrix} -\frac{\mathrm{D}}{2*10^3} - \frac{J}{2*10^3} - \frac{1}{5} & -\frac{N}{2*10^3} & -\frac{N}{2*10^3} & \frac{3\,\mathrm{I}}{25\,\sigma_2} - \frac{3\,\mathrm{I}}{50\,{\sigma_2}^2} - \frac{3\,\mathrm{I}^3}{10^6\,{\sigma_2}^2} & 0 & 0 \\ -\frac{\mathrm{D}}{2*10^3} & \sigma_4 & 0 & -\frac{\mathrm{I}\left(\frac{1}{\sigma_1} + \frac{V^2}{2*10^4\,\sigma_1}\right)}{20\,{\sigma_2}^2} & -\frac{10^3\left(\frac{V}{2*10^4\,{\sigma_1}^2} - \frac{V}{10^4\,\sigma_1} + \frac{V^3}{4*10^8\,{\sigma_1}^2}\right)}{\sigma_2} & 0 \\ -\frac{J}{2*10^3} & 0 & \sigma_4 & \frac{\mathrm{I}^4}{4*10^7\,\sigma_3} - \frac{\mathrm{I}^4}{8*10^7\,{\sigma_3}^2} - \frac{\mathrm{I}^9}{128*10^{17}\,{\sigma_3}^2} & 0 & 0 \\ \frac{1}{10} & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{20} \\ 0 & 0 & 0 & -\frac{\mathrm{I}}{20\,{\sigma_2}^2} & 0 & -\frac{3}{20} \end{bmatrix}$$

where $\sigma_1 = \dfrac{V^2}{4*10^4} + 1, \sigma_2 = \dfrac{I^2}{4*10^4} + 1, \sigma_3 = \dfrac{I^5}{32*10^{10}} + 1$ and $\sigma_4 = -\dfrac{N}{2*10^3} - \dfrac{3}{20}$.

After substituting the equilibrium values, presented by $\vec{x_1}, \vec{x_2}$ and $\vec{x_3}$, in the Jacobian matrix, the following eigenvalues were found:

$$\vec{x_1}(\lambda) = \begin{bmatrix} -4,8780 \\ -0,0838 \\ -0,1610 \\ -0,5243 \\ -0,4722 \\ -0,2871 \end{bmatrix}, \vec{x_2}(\lambda) = \begin{bmatrix} -2,5054 \\ 0,0217 \\ -0,5911 \\ -0,2325 \\ -0,3719 \\ -0,6005 \end{bmatrix}, \vec{x_3}(\lambda) = \begin{bmatrix} -2,5035 \\ -0,1109 \\ -0,1623 \\ -0,5808 \\ -0,4922 \\ -2,1045 \end{bmatrix}$$

In this sense, the dominant eigenvalue of $\vec{x_1}(\lambda)$ is $-0.0838 < 0$ and of $\vec{x_3}(\lambda)$ is $-0.1109 < 0$. Therefore, these are stable equilibrium points.

In $\vec{x_2}(\lambda)$ the dominant eigenvalue is $0.0217 > 0$. In this case, the equilibrium is unstable and, furthermore, as the other eigenvalues are less than zero, this is a saddle point.

By analyzing the values achieved for each protein at point T, it was observed that the stable equilibrium point T has a high concentration of *Delta* binding protein and VEGFR2 receptor protein. In addition, there are low concentrations of *Notch* receptor protein and *Jagged* binding protein at the T spot. Therefore, it can be concluded that this spot configures *tip* cells, which are the cells that will guide the growth of new blood vessels, corroborating the results obtained in [5].

When verifying the concentrations of proteins present at the stable equilibrium point S, high concentrations of *Notch* receptor protein, *Jagged* binding protein, and NICD were found. In addition, low concentrations of *Delta* binding protein, VEGFR2 receptor protein, and VEGF were observed. Therefore, point S has characteristics of *stalk* cells, which are the cells that will compose and lengthen the walls of blood vessels, corroborating the results obtained in [5].

Using this model and the parameters adopted in [5] it was possible to obtain consistent results even without using the *fringe* effect. Therefore, the model proposed here is considered functional and simpler than the one suggested by Boareto and collaborators [5].

Finally, the results presented in Figure 3 deserve to be highlighted, as there are fixed points that describe the states that will be seen later in the study by Alves [14].

5.2. The Influence of VEGF-A in the Model

When cells need oxygen and nutrients, chemical factors such as VEGF-A are released to signal this deficiency. In response to this signaling, the *tip* cell migrates towards the VEGF-A ligand guiding the formation of the new blood vessel that will carry oxygen and nutrients to these cells [5, 6].

The study presented by Alves [14] experimentally shows the role of VEGF-A inhibition in the vascular network growth process. However, it is important to mathematically understand the influence of this factor on the angiogenesis process.

By varying the value of VEGF-A (V_{ext}) in the model and keeping the other parameters, we have the Figure 4:

Figure 4 characterizes the plot of VEGFR2 versus VEGF-A, in which the values of VEGF-A were varied from 800 to 3000 molecules, in intervals of

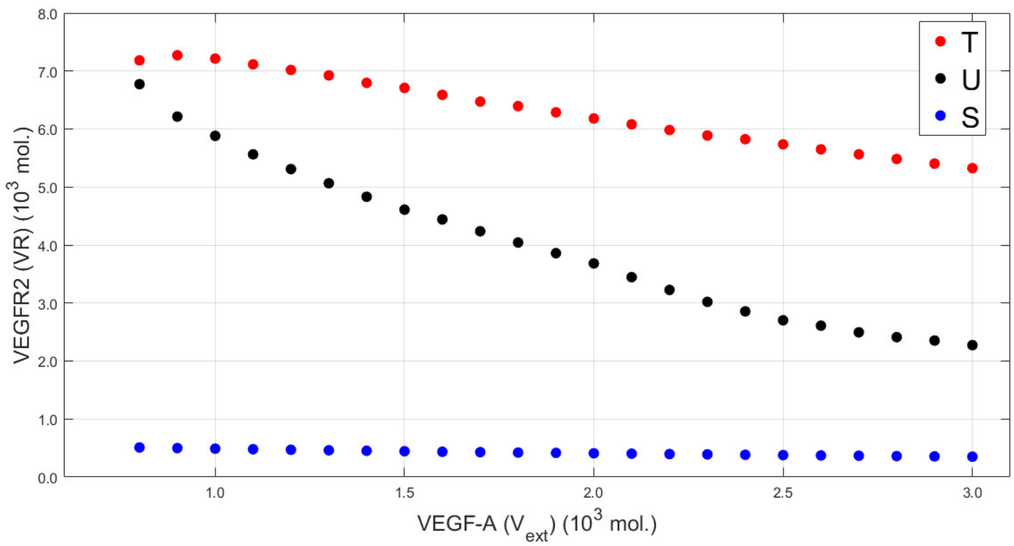

Figure 4: Phase representation. The new values of VEGFR2 to tip, U, and *stalk* equilibrium points of the model are given by the red, black and blue circles, respectively. The dotted line represented the data with VEGF-A = 2000 molecules.

100 molecules. Each point represented in Figure 4 presents new equilibrium concentrations for the model proteins.

For each new VEGF-A value, three new coordinates were obtained for the three equilibrium points. These new points referring to new VEGF-A concentrations maintained the same characteristics observed in Figure 3 (unstable U, stable T and stable S). Therefore, the analysis showed that the points called *tip* and *stalk* are stable and represent *tip* and *stalk* cells, respectively. Furthermore, the equilibrium point U is unstable, sometimes tending to the *stalk* point and sometimes tending to the *tip* point. This shows that the same characteristics were maintained when taking $V_{ext} = 2000$ molecules. However, the protein concentrations in each system were altered.

In Figure 4, it is observed that the higher the value of VEGF-A, the lower the concentration of VEGFR2 in the system. This fact is explained by the junction between VEGF-A and VEGFR2, forming the key that makes the receptor active, decreasing the concentration of free VEGFR2. Furthermore, according to [5], when VEGF-A+VEGFR2 is formed, the levels of *Delta* are increased in the system.

Figure 4 shows that the opposite is also true, because the lower the concentration of free VEGF-A, the greater the agglomeration of VEGFR2, since the compound VEGF-A+VEGFR2 will not be formed and the receptor will not be activated, inhibiting angiogenesis. By relating these dynamics to the introduction of the drug Avastin® in the system, which is an angiogenesis inhibitor drug, it is known that Avastin® receives VEGF-A, with the concentration of this ligand being decreased, which makes the concentration of the free VEGFR2 receptor higher.

Mathematical analysis showed that the increase and decrease of VEGF-A does not significantly alter the concentration of VEGFR2 at the *stalk* equilibrium point. The concentrations of all proteins in the model remains practically constant.

Another analysis that can be taken from Figure 4 is that the decrease in VEGF-A causes the concentrations of VEGFR2 at the *tip* and I point the approach, indicating that physiological angiogenesis can be achieved when free VEGF in the extracellular matrix decreases, as expected.

5.3. Relationship between the influence of VEGF on angiogenesis: a quantitative study

Alves [14] observed the growth of the vascular network in chicken embryos divided into two distinct groups: the control group and the treated group. The control group was used to extract quantitative parameters and that means that in this group the levels of VEGF-A were maintained to enable a physiological system. In the treated group, the drug Avastin® was applied to decrease VEGF signaling and this means that the values of free VEGF-A in the extracellular matrix were decreased causing the VEGFR2 receptor to increase.

To describe the differences between the network topologies, the following analyzes will present the behavior of the system in the control group and the group treated with Avastin®. The drug-treated the group represents the group in which VEGF-A was restricted.

One of the analyzes carried out in [14] was the verification of the length between the bifurcations of blood vessels in the two groups of embryos. When verifying this metric, Alves [14] uses the means obtained for the control and treated groups. This result is shown below in Figure 5

It can be seen from Figure 5a) that up to about 7.8 hours the average length between the bifurcations in the two groups grow similarly. From 7.8 hours onwards, the mean length between blood vessel bifurcations in the treated group is shorter than in the control group. This result is an indication that the action of Avastin® approaches the bifurcations in the treated

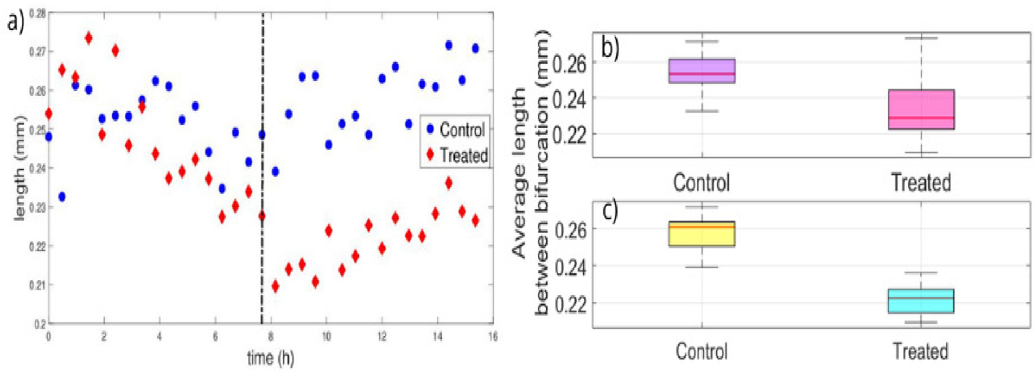

Figure 5: The average length between bifurcations for the control and treated groups. a) The average length between bifurcations, given in millimeters, in relation to time, given in hours. The blue circle represents the control group and the red square represents the group treated with Avastin®. The dotted line highlights the experiment time of 7.8 hours. b) represents the boxplots for the average length between bifurcations during 15 hours of the experiment and (c) represents the missing data between 7.8 and 15 hours of the experiment. The data were provided by the study by Alves[14].

group. In addition, [14] identify that these differences between the control and treated groups are more expressive after 7.8 hours of the experiment.

To analyze the relative position between the groups and check whether the trend presented after 7.8 hours is statistically significant, the boxplot of the two groups was constructed together. For this, the interval from 0 to 15 hours of the experiment (Figure 5b) and the interval from 7.8 to 15 hours of the experiment were considered (Figure 5c).

According to Figure 5b), in the treated group the average length between bifurcations is shorter than in the control group. This difference becomes clearer when we analyze the data from 7.8 hours onwards. This suggests that Avastin® acts on the vascular network, bringing the nodules closer to the blood vessels. However, in the control group, VEGF-A microdomains serve as a guide for creating pathways that direct cell migration.

During the blood vessel growth process, VEGF-A binds to Avastin®, inhibiting VEGFR2. This indicates that this system in the treated group is moving towards the equilibrium point concentrations U represented in Figure 3, indicating instability. However, it is worth highlighting that the decrease in VEGF-A due to the action of the drug induces the unstable system to move towards physiological angiogenesis with characteristics of *tip* cells, as represented in Figure 4.

On the other hand, it is noted that the blood vessels in the control group have longer lengths between the bifurcations. This indicates that the lumen of the forming vessel is more elongated than in the treated group, as a result of the lateral induction process. Therefore, it can be said that the control group presents characteristics of *stalk* cells and tends to the stable equilibrium point of the *stalk* represented in Figure 3.

When cutting from 7.8 hours onwards, this behavior is clearer, as the boxplots remain without intersection and clearly, the average length between bifurcations is shorter in the treated group, confirming the trend shown in Figure 5a).

It is important to consider Figure 5c) because according to Alves [14] the largest connected cluster is formed after 5 hours of the experiment. Therefore, after 7.8 hours of the experiment, the angiogenesis process is considered and no longer the vasculogenesis process (which is the initial formation of blood vessels in embryos).

This highlights that it was possible to find a real solution for the equilibrium points of the mathematical model presented.

Another analysis carried out in [14] refers to the average degree of the vertices as a function of time, that is, the number of connectivity of the nodes of the sample vessels, for the control and treated groups. The Figure 6 shows this representation.

The dotted line in Figure 6 a) is a reference to track the 5-hour time, which according to Alves [14] is when the largest connected cluster is formed. The black arrows identify the time of 7.8 hours, which is the time close to saturation for both groups. After 7.8 hours of experiment large differences are identified between the network topologies of the two groups studied [14].

The Figure 6 shows evidence of differences between the connectivity of the vascular network in the control and treated groups. This result corroborates the study presented in [46] which states that antiangiogenic drugs promote a reorganized vascular network to improve oxygenation and drug penetration into the system.

To analyze the relative position between the groups and check whether the trend presented in Figure 6a) after 7.8 hours is statistically significant, the two groups will be plotted in a boxplot together. For this, the interval from 0 to 15 hours of the experiment (Figure 6b) and the interval from 7.8 to 15 hours of the experiment (Figure 6c) will be considered.

The Figure 6b) presents the data referring to 15 hours of the experiment. There is an intersection between the boxes and a large scatter of data.

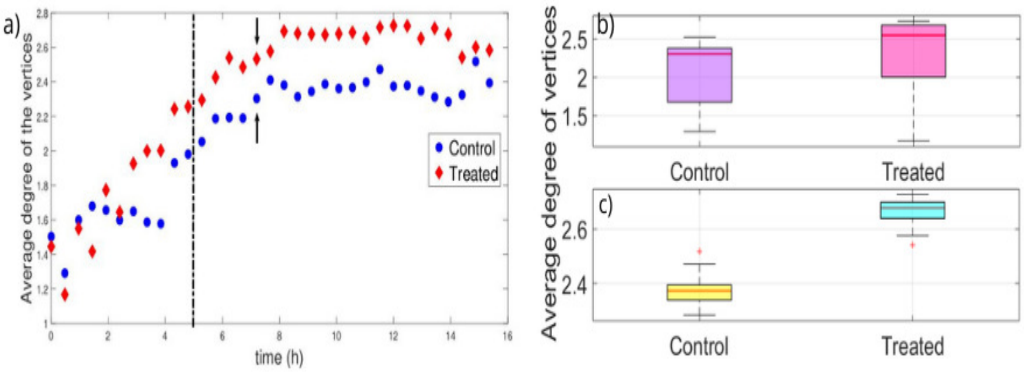

Figure 6: The average degree of the vertices for the control and treated groups. a) bifurcation growth behavior in the control and treated groups. The blue circle represents the control group and the red square represents the treated group. The dotted line indicates the time of 5 hours and the arrows highlight the experiment time of 7.8 hours. b) represents the boxplot for two groups with data from 15 hours of the experiment and c) represents the boxplot for two groups with data between 7.8 and 15 hours of the experiment The data were provided by the study by Alves[14].

However, the analysis after 7.8 hours of the experiment, in which data related to angiogenesis are effectively observed, shows independence between groups, symmetry and low dispersion in the data, as represented by Figure 6c).

It can be said that the treated group presents higher values for the average degree of the vertices. This shows that the trend shown in Figure 6a) from 7.8 hours onwards is statistically significant. According to Alves [14], Avastin® acts on the system making connections random, indicating the instability presented by the balance point that U represented in Figure 3. Furthermore, this greater connectivity capacity indicates that this unstable system is heading towards high concentrations of *Delta* and VEGFR2, which characterizes a tendency towards physiological angiogenesis given by the *tip* cells.

On the other hand, the lower connectivity in the control group shows that the concentrations of *Delta* and VEGFR2 in the system, which give rise to the *tip* cells that lead vessel growth, are lower than the concentrations of *Notch*, *Jagged* and NICD. Therefore, the control group system tends to the *stalk* equilibrium point that reflects a high incidence of *stalk* cells, as represented in Figure 3.

6. Conclusions

Understanding how new blood vessels grow according to signals emitted by signaling pathways is one of the essential factors in understanding angiogenesis and tumor development, and this field is an intense subject of research [49] .

When there is unrestrained growth of blood vessels designed to supply oxygen and nutrients to a tumor, it creates a favorable environment for tumor growth and metastasis [15].

In this study, a mathematical model was proposed to describe the cellular dynamics involved in the angiogenesis process, based on the models presented in [5] and [13].

To validate the proposed model, the steps highlighted in [13] were followed and the qualitative analyzes were similar, resulting in a simpler mathematical and functional model. The model presented here, and based on the model by Boareto and collaborators [5], satisfactorily describes the concentrations of receptor proteins and ligands involved in cellular dynamics throughout the angiogenesis process.

In this study, the *Notch* and VEGF signaling pathways and the proteins that act in these pathways to promote angiogenesis were presented and analyzed.

Dysregulated *Notch* signaling can facilitate the emergence and proliferation of cancer stem cells, activating factors that promote cell survival and pathological angiogenesis [13]. This signaling pathway influences some types of cancer, such as breast and prostate cancer [50, 51].

Likewise, VEGF signaling contributes to the self-renewal and survival of cancer stem cells and is also related to several types of cancer, such as bone cancer [52, 53].

Some studies such as [5], [54] and [51] indicate that high levels of *Jagged* proteins demonstrate characteristics associated with the emergence and maintenance of tumors. Overexpression of *Jagged* gives rise to a hybrid *tip/stalk* phenotype causing the fate of *tip* and *stalk* cells to be destabilized, making angiogenesis chaotic and generating excessive vessels inadequately perfused, which is one of the hallmarks of cancer. Furthermore, other studies, such as [52], suggest that VEGF-A also has a great influence on the development of cancer cell metastases. Therefore, the importance of research that discusses the influence of ligand and receptor proteins during angiogenesis is considered.

The model proposed in this study disregards the performance of the *fringe* effect, as the initial analysis of the study showed that the inclusion of this effect was not decisive in the mathematical modeling proposed in [5]. However, for new and different approaches, it is suggested to evaluate the performance of the *fringe* according to the proposed objective.

The results presented in this work satisfactorily present the cellular dynamics of a single cell, which is immersed in an external environment in which other external concentrations are simulated, as done in [5] and [13].

An important fact is that it was possible to correlate the system's equilibrium points with the [14] experiments, in which the interaction of cell clusters in forming vascular networks was studied. In this sense, the abstract mathematical model was applied to the experimental study by Alves[14], in which it was possible to explain the formation of the vascular network. This relationship was obtained using the model and varying the amount of VEGF-A, given VEGF signaling. Once this was done, it was noticed that the balance of the system was disturbed and generated changes in the blood vessel growth process. It was observed that the difference between the network topologies of the control group and the treated group became more evident after 7.8 hours of experiment.

The growth of new blood vessels in the control group observed by Alves [14] corresponds to a state of high concentration of stem cells, which make up the formation of vessel walls. However, the growth of new blood vessels in the treated group, given the reflections of the model, reflects a tendency towards equilibrium point I, representing instability. Therefore, it was possible to obtain a solution for ordinary differential equations with fixed points that corresponds to the results obtained by Alves[14]. Thus, it is highlighted that it was possible to find a real solution for the equilibrium points of the abstract model.

Just like healthy cells, malignant cells need oxygen and nutrients to survive and proliferate. These malignant cells reside close to blood vessels to more easily access the bloodstream [55]. On this occasion, *Notch*, *Delta*, *Jagged* and VEGF-A have important roles in the development and cancer progression and analyzes to inhibit, target and manipulate *Notch* and VEGF signaling pathways are important for generating new types of cancer treatments [50]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement

The original contributions presented in the study are included in the Appendix A, further inquiries can be directed to the corresponding author.

Appendix A. The Mathematical Model

The mathematical model presented here is a model adapted from Boareto and collaborators [13]. This model differs in that it does not add the performance of the *fringe* effect, which is an effect that models part of the production of *Notch* to make it more apt to receive *Delta* instead of *Jagged*. Qualitative analyzes without *fringe* inclusion were analogous to [13], making the model mathematically simpler and more functional.

Before talking about the model equations, it is important to talk about the shifted Hill equation, which is the equation that will represent the action of the NICD in increasing or restricting the rates of protein production in the cell nucleus. The shifted Hill equation is defined as:

$$H^S(X, \lambda_{X,Y}) = H^-(X) + \lambda_{X,Y} H^+(X) \tag{A.1}$$

where $H^{-}(X)$ is a constraint function, $H^{+}(X)$ is an activation function, and $\lambda_{X,Y}$ denotes the change in the output of Y according to X. The restriction function and the activation function in the shifted Hill function are related to the parameters to be used in the model. For the activation function (H^{S+}) we use $\lambda > 1$ and for the constraint function (H^{S-}) we have $\lambda = 0$.

The shifted Hill function referring to NICD signaling is:

$$H^{S}(I,\lambda) = \frac{1}{1+\left(\dfrac{I}{I_0}\right)^n} + \lambda \frac{\left(\dfrac{I}{I_0}\right)^n}{1+\left(\dfrac{I}{I_0}\right)^n} \tag{A.2}$$

where n is the Hill exponent, which assumes different values for each protein depending on activation or restriction due to the action of the NICD. The parameter λ is also a constant that depends on whether the Hill-shifted function is positive (H^{S+} = activation function $\lambda > 0$) or negative (H^{S-} = constraint function $\lambda = 0$). The term I represents the concentration of NICD and I_0 is a constant.

When considering the inclusion of VEGF in the system, there is a contribution in the rate of production of proteins textit Delta. *Delta* is restricted by receptor signaling *Notch*, but is activated by VEGF signaling.

To include this activation of *Delta*, which also occurs by the action of the NICD, the shifted Hill function is analogous to Equation A.2. Therefore, the Hill function shifted in the function of VEGF is:

$$H^{S}(V,\lambda) = \frac{1}{1+\left(\dfrac{V}{V_0}\right)^n} + \lambda \frac{\left(\dfrac{V}{V_0}\right)^n}{1+\left(\dfrac{V}{V_0}\right)^n} \tag{A.3}$$

The term V represents the concentration of VEGF and V_0 is a constant.

When considering Equation A.2 for *Notch* and *Jagged*, the model assumes the positive shifted Hill function ($\lambda > 0$) and when considering Equation A.2 for *Delta* and VEGFR2 the shifted Hill function is negative ($\lambda = 0$). However, when considering the Equation A.3, the shifted Hill function becomes positive for *Delta* ($\lambda > 0$), since signaling via VEGF regulates the increase of this protein [5].

The shifted Hill functions, represented by Equation A.2 and Equation A.3, are efficient to describe state variations, being also useful to model the production of substances when their origins are being regulated by others [56].

Appendix A.1. Mathematical Model Details

Appendix A.1.1. Construction of the Notch equation

First, the equation for *Notch* will be built. For this, one must consider the link between *Notch* of a cell (N) and *Delta* of an external environment (D_{ext}) and this involves the interaction *trans* (k_t). This dynamic is represented by the Equation A.4.

$$[N] + [D_{ext}] \longleftrightarrow [ND_{ext}] \tag{A.4}$$

Note that Equation A.4 occurs in two directions. k_{t-} will be rendered right to left and k_{t+} left to right.

One must also consider the four factors that determine the variation of the protein *Notch* in a cell, which are:

- The NICD: which penetrates the cell nucleus activating the production of *Notch*. This makes the population of *Notch* concerning time to be proportional to the rate of production of *Notch* (N_0) multiplied by the shifted Hill function. This relationship is represented by Equation A.5, where $\lambda_{I,N}$ is the parameter represented by λ in Equation A.2 for the protein *Notch* and is related to the positive Hill function displaced by the NICD performance.

$$\frac{dN}{dt} \propto +N_0 H^{S+}(I, \lambda_{I,N}) \tag{A.5}$$

- k_{t+}: if the Equation A.4 happens in the direction of k_{t+}, there is a decrease in population of N (and of D_{ext}) , which binds to D_{ext} to form another compound that also depends on D_{ext}. This relationship is represented by:

$$\frac{dN}{dt} \propto -k_{t+} N D_{ext} \tag{A.6}$$

- k_{t-}: if the Equation A.4 happens in the direction of k_{t-}, there is a population increase of N (and of D_{ext}):

$$\frac{dN}{dt} \propto +k_{t-} [N D_{ext}] \tag{A.7}$$

- Natural degradation: the population of *Notch* decreases by a process of natural degradation that occurs at a rate γ:

$$\frac{dN}{dt} \propto -\gamma N \tag{A.8}$$

However, the inhibition *cis* can occur between *Notch* of the considered cell and *Delta* of the same cell. Like the interaction *trans*, the inhibition *cis* has different rates (k_{c+} and k_{c-}) and is represented by:

$$[N] + [D] \longleftrightarrow [ND] \tag{A.9}$$

According to Equation A.9, there are two more terms that will contribute $\frac{dN}{dt}$, which are:

- k_{c+}: is when the Equation A.9 happens from left to right. In this case, there will be a decrease in *Notch*, as *Notch* and *Delta* interact to form $[ND]$. This dynamic is represented by:

$$\frac{dN}{dt} \propto -k_{c+} N D \tag{A.10}$$

- k_{c-}: is when the Equation A.9 happens in a right-to-left direction. In this case, there will be an increase of *Notch* as $[ND]$ breaks. This dynamic is represented by:

$$\frac{dN}{dt} \propto +k_{c+} [ND] \tag{A.11}$$

Putting together all the equations that represent $\frac{dN}{dt}$, so far, we have:

$$\frac{dN}{dt} = N_0 H^{S+}(I, \lambda_{I,N}) - N(k_{t+}D_{ext} + k_{c+}D) + k_{t-}[ND_{ext}] + k_{c-}[ND] - \gamma N \tag{A.12}$$

Equations representing the variations of $[ND_{ext}]$ and $[ND]$ must also be considered.

For $[ND_{ext}]$:

- The production of this compound is increased at a rate k_{t+}, as represented by Equation A.4, and depends on the concentrations of $[N]$ and $[D_{ext}]$:

$$\frac{d[ND_{ext}]}{dt} \propto +k_{t+}ND_{ext} \tag{A.13}$$

- However, the output of $[ND_{ext}]$ is reduced by Equation A.4 when it happens from right to left:

$$\frac{d[ND_{ext}]}{dt} \propto -k_{t-}[ND_{ext}] \tag{A.14}$$

- there is also the degradation rate (k_l) of this compound in relation to a signal sent by the NICD:

$$\frac{d[ND_{ext}]}{dt} \propto -k_l[ND_{ext}] \tag{A.15}$$

Bringing together the Equation A.13, Equation A.14 and Equation A.15, we have:

$$\frac{d[ND_{ext}]}{dt} = +k_{t+}ND_{ext} - k_{t-}[ND_{ext}] - k_l[ND_{ext}] \tag{A.16}$$

To [ND]:

By arguments similar to those used to construct the equation of $[ND_{ext}]$, the variation of $[ND]$ is given by:

$$\frac{d[ND]}{dt} = +k_{c+}ND - k_{c-}[ND] - k_l[ND] \tag{A.17}$$

Assuming that these populations reach an equilibrium state (steady state), that is, $\dfrac{d[ND_{ext}]}{dt} = 0$ and $\dfrac{d[ND]}{dt} = 0$:

$$[ND_{ext}] = \frac{k_{t+}}{k_{t-} + k_l} ND_{ext} \tag{A.18}$$

$$[ND] = \frac{k_{c+}}{k_{c-} + k_l} ND \tag{A.19}$$

Replacing Equation A.18 and Equation A.19 into Equation A.12 and performing some mathematical manipulations, we have:

$$\frac{dN}{dt} = N_0 H^{S+}(I, \lambda_{I,N}) - k_{t+}ND_{ext}\left(\frac{k_l}{k_{t-} + k_l}\right) - k_{c+}ND\left(\frac{k_l}{k_{t-} + k_l}\right) - \gamma N \tag{A.20}$$

Defining $k_t = \dfrac{k_{t+}k_l}{k_{t-} + k_l}$ e $k_c = \dfrac{k_{c+}k_l}{k_{c-} + k_l}$ the equation from textit Notch will be:

$$\frac{dN}{dt} = N_0 H^{S+}(I, \lambda_{I,N}) - k_t ND_{ext} - k_c ND - \gamma N \tag{A.21}$$

When *Notch* of the cell being analyzed (N) receives *Jagged* from the external environment (J_{ext}), the activation *trans* (k_t) and the NICD penetrates the nucleus of the cell that contains *Notch* activating the production of these two proteins. However, when the cell's *Notch* receives *Jagged* (J) from the same cell, the NICD is not triggered and both proteins are degraded by the *cis* (k_c) inhibition process. Analogous to the observations made when considering the dynamics between *Notch* and *Delta*, by adding *Jagged* in the Equation A.21, we have:

$$\frac{dN}{dt} = N_0 H^{S+}(I, \lambda_{I,N}) - k_t N(D_{ext} + J_{ext}) - k_c N(D + J) - \gamma N \tag{A.22}$$

As the inclusion of VEGF does not directly affect the production of Notch, the mathematical model that represents *Notch* is given by Equation A.22.

Appendix A.1.2. Construction of the Delta equation

Now, the variation *Delta* of the cell in question will be modeled.

The protein *Delta* (D) of this cell can interact with *Notch* from an external environment (N_{ext}), analogous to Equation A.4.

$$[N_{ext}] + [D] \longleftrightarrow [N_{ext}D] \tag{A.23}$$

If the reaction of Equation A.23 occurs from left to right (k_{t+}), *Delta* decreases, but in the reverse direction (k_{t-}), *Delta* increases:

$$\frac{dD}{dt} \propto -k_{t+} N_{ext} D + k_{t-}[N_{ext}D] \tag{A.24}$$

The formation of NICD (I) restricts the production of *Delta*, so one must multiply the production rate of *Delta* (D_0) by the "negative" shifted Hill function, ie, per:

$$H^{S-}(I, \lambda_{I,D}) = \frac{1}{1 + \left(\dfrac{I}{I_0}\right)^n} \tag{A.25}$$

resulting in:

$$\frac{dD}{dt} \propto D_0 H^{S-}(I, \lambda_{I,D}) \tag{A.26}$$

One must also consider the natural degradation rate (γ) of *Delta*:

$$\frac{dD}{dt} \propto -\gamma D \tag{A.27}$$

In addition, the inhibition *cis* can occur between *Notch* and *Delta* of the same analyzed cell, like this:

$$\frac{dD}{dt} \propto -k_{c+}ND + k_{c-}[ND] \tag{A.28}$$

Putting together all the equations that represent $\frac{dD}{dt}$, so far, we have:

$$\frac{dD}{dt} = D_0 H^{S-}(I, \lambda_{I,D}) - k_{t+}N_{ext}D - k_{c+}ND + k_{t-}[N_{ext}D] + k_{c-}[ND] - \gamma D \tag{A.29}$$

The equations representing the variations of $[N_{ext}D]$ and $[ND]$ must be considered, as was done for *Notch*.

When reviewing the arguments leading to Equation A.16, Equation A.17, Equation A.18 and Equation A.19, the Equation A.16 will be changed to:

$$\frac{d[N_{ext}D]}{dt} = +k_{t+}N_{ext}D - k_{t-}[N_{ext}D] - k_l[N_{ext}D] \tag{A.30}$$

and, in the steady state, that is, $\frac{d[N_{ext}D]}{dt} = 0$, there will be an equation analogous to Equation A.18, represented by:

$$[N_{ext}D] = \frac{k_{t+}}{k_{t-} + k_l} N_{ext}D \tag{A.31}$$

For $[ND]$, Equation A.19 remains.

Replacing Equation A.31 and Equation A.19 into Equation A.29, the model is given by:

$$\frac{dD}{dt} = D_0 H^{S-}(I, \lambda_{I,D}) - k_t N_{ext}D - k_c ND - \gamma D \tag{A.32}$$

As already seen, it should be considered that the *Delta* production rate (D_0) is restricted by the NICD's performance by the *Notch* signaling pathway and increased by the NICD's effect by the VEGF signaling pathway. Therefore, to model the inclusion of VEGF performance in the concentration of *Delta*, just insert the term $H^+(V, \lambda_{V,D})$ by multiplying the production rate of *Delta* (D_0) in Equation A.32. Thus, the mathematical model for *Delta* will be:

$$\frac{dD}{dt} = D_0 H^{S-}(I, \lambda_{I,D}) H^{S+}(V, \lambda_{V,D}) - k_t N_{ext} D - k_c N D - \gamma D \quad \text{(A.33)}$$

Appendix A.1.3. Construction of the Jagged equation

Now, the *Jagged* variation of the cell being considered will be modeled. This analysis will be similar to the analysis performed for *Delta*.

The *Jagged* (J) protein of this cell can interact with *Notch* from an external environment (N_{ext}), so:

$$[N_{ext}] + [J] \longleftrightarrow [N_{ext}J] \quad \text{(A.34)}$$

If the reaction occurs from left to right (k_{t+}), *Jagged* int, but in reverse (k_{t-}), *Jagged* increases:

$$\frac{dJ}{dt} \propto -k_{t+} N_{ext} J + k_{t-} [N_{ext} J] \quad \text{(A.35)}$$

The formation of NICD (I) activates the production of *Jagged*, so one must multiply a production taxon of *Jagged* (J_0) by the shifted Hill function:

where $\lambda_{I,J}$ is the parameter represented by λ in Equation A.2 and is related to *Jagged*. One must also consider the natural degradation rate (γ) of *Jagged*.

$$\frac{dJ}{dt} \propto -\gamma J \quad \text{(A.36)}$$

In addition, *cis* inhibition may occur between *Notch* and *Jagged* of the same analyzed cell:

$$\frac{dJ}{dt} \propto -k_{c+}NJ + k_{c-}[NJ] \tag{A.37}$$

By putting together all the equations that represent $\frac{dJ}{dt}$, so far, we have:

$$\frac{dJ}{dt} = J_0 H^{S+}(I, \lambda_{I,J}) - k_{t+}N_{ext}J - k_{c+}NJ + k_{t-}[N_{ext}J] + k_{c-}[NJ] - \gamma J \tag{A.38}$$

Now, one must consider the equations that represent the variations of $[N_{ext}J]$ and $[NJ]$, as was done for *Notch* and *Delta*.

When reviewing the arguments leading to Equation A.16, Equation A.17, Equation A.18 and Equation A.19, the Equation A.30 will be given changed to:

$$\frac{d[N_{ext}J]}{dt} = +k_{t+}N_{ext}J - k_{t-}[NextJ] - k_l[N_{ext}J] \tag{A.39}$$

and, in the steady state, that is, $\frac{d[N_{ext}J]}{dt} = 0$, there will be an equation analogous to Equation A.31, resulting in:

$$[N_{ext}J] = \frac{k_{t+}}{k_{t-} + k_l} N_{ext}J \tag{A.40}$$

For $[NJ]$, analogously to Equation A.40, we arrive at:

$$[NJ] = \frac{k_{c+}}{k_{c-} + k_l} NJ \tag{A.41}$$

By replacing Equation A.40 and Equation A.41 in Equation A.38, the mathematical model for *Jagged* will be given by:

$$\frac{dJ}{dt} = J_0 H^{S+}(I, \lambda_{I,J}) - k_t N_{ext}J - k_c NJ - \gamma J \tag{A.42}$$

Appendix A.1.4. Construction of the NICD equation

For the NICD mathematical model, one must consider that a NICD output (I) affects *Notch*, *Delta* and *Jagged*. The NICD is only produced by the interaction textit trans between textit Notch of the analyzed cell and textit Delta and textit Jagged of the external environment (D_{ext} e J_{ext}):

$$\frac{dI}{dt} \propto +k_{t+}ND_{ext} - k_{t-}[ND_{ext}] + k_{t+}NJ_{ext} - k_{t-}[NJ_{ext}] \tag{A.43}$$

There is also the NICD degradation rate which will be given by γ_I:

$$\frac{dI}{dt} \propto +k_{t+}ND_{ext} - k_{t-}[ND_{ext}] + k_{t+}NJ_{ext} - k_{t-}[NJ_{ext}] - \gamma_I I \tag{A.44}$$

Since $[ND_{ext}]$ is given by Equation A.18, and by similar arguments $[NJ_{ext}]$ is found, in steady state, when performing some mathematical manipulations, the equation representing the NICD is:

$$\frac{dI}{dt} = k_t N(D_{ext} + J_{ext}) - \gamma_I I \tag{A.45}$$

Appendix A.1.5. Construction of the VEGF equation

Now, VEGF production will be considered. This factor, like the NICD, is only produced by the interaction *trans* (k_t) which in this case will occur when the cell's VEGFR2 receptor (VR) receives the VEGF-A ligand (V_{ext}):

$$\frac{dV}{dt} \propto +k_{t+}V_{ext}VR - k_{t-}[V_{ext}VR] \tag{A.46}$$

There is also the VEGF degradation rate that will be given by γ_I, so, incorporating this analysis, we have:

$$\frac{dV}{dt} \propto +k_{t+}V_{ext}VR - k_{t-}[V_{ext}VR] - \gamma_I V \tag{A.47}$$

By arguments similar to those used to determine $[ND_{ext}]$, we find the equation of the term $[V_{ext}VR]$ and performing some mathematical manipulations, the equation that represents the dynamics of VEGF will be given by:

$$\frac{dV}{dt} = k_t V_{ext} VR - \gamma_I V \tag{A.48}$$

Appendix A.1.6. Construction of the VEGFR2 equation

To model VEGFR2, it must be considered that its production rate (VR_0) is inhibited by the NICD, therefore:

$$\frac{dVR}{dt} \propto VR_0 H^{S-}(I, \lambda_{I,VR}) \tag{A.49}$$

There is also the natural degradation rate of VEGFR2 (γ):

$$\frac{dVR}{dt} \propto -\gamma VR \tag{A.50}$$

Furthermore, this reaction only happens upon activation *trans* (k_t):

$$\frac{dVR}{dt} \propto -k_{t+} V_{ext} VR + k_{t-}[V_{ext}VR] \tag{A.51}$$

Combining the Equation A.49, Equation A.50 and Equation A.51, we have:

$$\frac{dVR}{dt} = VR_0 H^{S-}(I, \lambda_{I,VR}) - k_{t+} V_{ext} VR + k_{t-}[V_{ext}VR] - \gamma VR \tag{A.52}$$

Considering the equation that represents the variation of $[V_{ext}VR]$, by arguments similar to those used for the *Notch* signaling, we arrive at Equation A.53, which represents the variation of the VEGFR2.

$$\frac{dVR}{dt} = VR_0 H^{S-}(I, \lambda_{I,VR}) - k_t V_{ext} VR - \gamma VR \tag{A.53}$$

Appendix A.2. Parameters used

The parameters were taken from the studies of [5]. The Table A.1 and Table A.2 represents the values used.

Table A.1: Parameters used for the system composed of the equations for Notch, Delta, Jagged and NICD proteins

Parameter	Value	Unit of Measurement
N_0	1600	Number of proteins / time (h)
J_0	1200	Number of proteins / time (h)
D_0	1800	Number of proteins / time (h)
I_0	200	Number of proteins / time (h)
N_{ext}	500	Number of proteins
J_{ext}	1750	Number of proteins
D_{ext}	0	Number of proteins
γ	0.1	$time^{-1}(h^{-1})$
γ_I	0.5	$time^{-1}(h^{-1})$
$\lambda_{I,N} = \lambda_{I,J}$	2	Dimensionless
$\lambda_{I,D}$	0	Dimensionless
$n_D = n_N$	2	Dimensionless
n_J	5	Dimensionless
k_t	$5 * 10^{-5}$	$time^{-1}(h^{-1})$
k_c	$5 * 10^{-4}$	$time^{-1}(h^{-1})$

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Table A.2: Parameters used for the system composed of the equations for Notch, Delta, Jagged, NICD, VEGF-A and VEGFR2 proteins

Parameter	Value	Unit of Measurement
N_0	1200	Number of proteins / time (h)
J_0	800	Number of proteins / time (h)
$D_0 = VR_0$	1000	Number of proteins / time (h)
$I_0 = V_0$	200	Number of proteins / time (h)
$N_{ext} = D_{ext} = J_{ext} = V_{ext}$	2000	Number of proteins
γ	0.1	$time^{-1}(h^{-1})$
γ_I	0.5	$time^{-1}(h^{-1})$
$\lambda_{I,N} = \lambda_{I,J} = \lambda_{V,D}$	2	Dimensionless
$\lambda_{I,D} = \lambda_{I,VR}$	0	Dimensionless
$n_D = n_N = n_V = n_{VR}$	2	Dimensionless
n_J	5	Dimensionless
k_t	$2.5 * 10^{-5}$	$time^{-1}(h^{-1})$
k_c	$5 * 10^{-4}$	$time^{-1}(h^{-1})$

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